SDE Discretisation: Itô-Taylor Expansions, Euler–Maruyama, and Milstein Schemes

Quant Finance Portfolio — Phase 2, Block 1, Algorithm 2 (theory)

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Abstract

The exact GBM sampler used in Block 1.1 exploited a closed-form solution of the underlying SDE. Beyond Black–Scholes, closed forms are essentially unavailable: pricing under local volatility, stochastic volatility, jump-diffusion, or path-dependent payoffs all require *simulating* approximate trajectories of the underlying stochastic process by discretising the SDE in time. This document develops the foundations of that discretisation: the two notions of convergence relevant to SDEs (*strong* and *weak*), the Itô-Taylor expansion that generates discretisation schemes systematically, and the two canonical schemes of order one in the relevant senses (Euler–Maruyama and Milstein). The emphasis is conceptual: the precise statements of convergence theorems are given without proof (Kloeden–Platen is the canonical reference), but the *reason* behind each convergence order is developed in detail through the Itô-Taylor expansion, and the practical consequences for finance applications are spelled out.

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1 Introduction

Block 1.1 priced the European call by drawing S_T from its exact distribution under geometric Brownian motion. The construction was trivial because the GBM SDE

$$\mathrm{d}S_t \;=\; rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t \tag{1}$$

admits a closed-form solution, $S_T = S_0 \exp((r - \sigma^2/2)T + \sigma W_T)$, so a single normal draw suffices to generate one path's terminal value. The exactness was a luxury, not a method.

Almost every model of practical interest beyond Black–Scholes lacks such a closed form. Local volatility, where σ depends on t and S_t, has no closed form for S_T even in simple parameterisations. Stochastic volatility models like Heston have a known characteristic function but no closed form for the joint (S_T, V_T). Jump-diffusion models add a Poisson component that destroys the log-normal structure entirely. In all these cases, the only practical route to $\mathbb{E}[f(S_T)]$ is to simulate *approximate* trajectories by discretising the SDE in time and then averaging.

The discretisation introduces a new kind of error, qualitatively different from the statistical error of Block 1.1. The Monte Carlo estimator was unbiased: any deviation between $\hat{C}_n$ and C_{BS} was purely sampling error, controlled by $1/\sqrt{n}$. Once we replace the exact S_T by a discretised $\hat{S}_T$, the estimator becomes *biased*: even with $n = \infty$ paths, we estimate $\mathbb{E}[f(\hat{S}_T)]$, not $\mathbb{E}[f(S_T)]$. The size of the bias depends on the time step h and on the scheme, and the central analytical question of this block is: at what rate does this bias vanish as $h \to 0$?

The answer is not a single number. SDE solutions admit two distinct and largely independent notions of approximation, called *strong* and *weak* convergence. The strong notion measures pathwise proximity, the weak notion measures distributional proximity, and a discretisation scheme can be of different order in the two senses. The most consequential example is the Euler–Maruyama scheme, which has strong order $\frac{1}{2}$ but weak order 1.

The remainder of this document is structured as follows. Section 2 fixes the SDE setting and introduces the abstract notion of a discretisation scheme. Section 3 develops the two notions of convergence in parallel, with examples and an explicit relation between them. Section 4 derives the Itô-Taylor expansion at the level of detail required to read off discretisation schemes from it. Section 5 derives the Euler–Maruyama scheme as a first-order truncation and states

its convergence orders. Section 6 does the same for the Milstein scheme as a second-order truncation. Section 7 discusses which scheme is appropriate in which context. Section 8 sets up the implementation work of Blocks 1.2.1 and 1.2.2.

2 Setting and discretisation schemes

2.1 The scalar Itô SDE

We work with a scalar Itô SDE

$$\mathrm{d}X_t \;=\; a(X_t)\,\mathrm{d}t \;+\; b(X_t)\,\mathrm{d}W_t, \qquad X_0 = x_0, \qquad t \in [0,T], \tag{2}$$

on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in[0,T]}, \mathbb{P})$ supporting a standard Brownian motion W. The coefficients $a, b : \mathbb{R} \to \mathbb{R}$ are taken sufficiently regular (Lipschitz, with linear growth) for a unique strong solution to exist; we will state the additional regularity required by each convergence theorem when needed.

Geometric Brownian motion is the running example: $a(x) = rx$, $b(x) = \sigma x$. We will refer back to it both for sanity checks and because the closed-form solution of GBM provides the reference path against which the discretised paths of Blocks 1.2.1 and 1.2.2 will be validated empirically.

2.2 Discretisation schemes

Fix a partition

$$0 = t_0 < t_1 < \cdots < t_N = T, \qquad t_n = nh, \qquad h = T/N, \tag{3}$$

and the corresponding Brownian increments

$$\Delta W_n \;\stackrel{\text{def}}{=}\; W_{t_{n+1}} - W_{t_n} \;\sim\; \mathcal{N}(0,h), \qquad n = 0,1,\ldots,N-1. \tag{4}$$

The increments ΔW_n are independent across n by the independent-increments property of Brownian motion.

A *discretisation scheme* for (2) is a deterministic recursion of the form

$$\hat{X}_{n+1} \;=\; \Psi\Big(\hat{X}_n,\; h,\; \Delta W_n,\; \ldots\Big), \qquad \hat{X}_0 = x_0, \tag{5}$$

where Ψ is some function of the current state, the step size, the Brownian increment, and possibly higher-order random variables (we will see exactly which in the Itô-Taylor section). The output is a sequence $\hat{X}_0, \hat{X}_1, \ldots, \hat{X}_N$ that we interpret as approximations to $X_{t_0}, X_{t_1}, \ldots, X_{t_N}$.

Two clarifications. First, the partition is uniform throughout this document. Adaptive step sizes exist and have their place but are not needed here. Second, the scheme is deterministic given $\Delta W_0, \ldots, \Delta W_{N-1}$: same Brownian path, same scheme, same output, no internal randomness in Ψ beyond the increments.

3 Strong and weak convergence

The fundamental distinction of SDE numerical analysis, absent from the deterministic ODE setting, is that there is more than one natural sense in which $\hat{X}_N$ approximates X_T. We develop the two relevant notions and explore their relationship.

3.1 Pathwise vs distributional approximation

Suppose we run the scheme with the *same* Brownian increments ΔW_n that were used to build the exact solution X_T on a common probability space. Two natural questions about the quality of the approximation suggest themselves:

1. How close is $\hat{X}_N$ to X_T, *path by path*? That is, on a typical realisation $\omega \in \Omega$, how large is $|X_T(\omega) - \hat{X}_N(\omega)|$?
2. How close is the distribution of $\hat{X}_N$ to that of X_T? That is, for nice test functions f, how large is $|\mathbb{E}[f(\hat{X}_N)] - \mathbb{E}[f(X_T)]|$?

These questions are formally distinct. The first requires that X_T and $\hat{X}_N$ live on the same probability space, which we arrange by sharing the Brownian path. The second is about marginal distributions and does not require any coupling: even if $\hat{X}_N$ were generated from an entirely separate Brownian motion, the expectation $\mathbb{E}[f(\hat{X}_N)]$ is well-defined and the comparison with $\mathbb{E}[f(X_T)]$ remains sensible.

The first question is what *strong* convergence formalises; the second, what *weak* convergence formalises.

3.2 Strong convergence

Definition 3.1 (Strong convergence)**.** A discretisation scheme is said to converge *strongly* to the solution of (2) with order $\gamma > 0$ if there exist constants $C > 0$ and $h_0 > 0$ such that, when the scheme is run on the Brownian path of X,

$$\mathbb{E}\Big[\,|X_T - \hat{X}_N|\,\Big] \;\le\; C\,h^{\gamma} \qquad \text{for all } h \in (0, h_0]. \tag{6}$$

The expectation is taken over the (shared) Brownian path. The definition uses L^1 norm, which is the most common choice; some references use L^2 instead, with a corresponding adjustment of the constant C but the same order γ for all the schemes we will consider.

The strong error (6) measures average pathwise deviation. It is the relevant quantity whenever a single trajectory of the discretised process is meant to stand in for a single trajectory of the true process: in pathwise sensitivity analysis (greeks via the pathwise method, see Block 4), in optimal control problems where the control is adapted to the simulated path, and in any context where two related quantities must be evaluated on the *same* simulated path.

3.3 Weak convergence

Definition 3.2 (Weak convergence)**.** A discretisation scheme is said to converge *weakly* to the solution of (2) with order $\beta > 0$ if for every test function f in a specified class of smooth functions (typically polynomials of degree $\le 2\beta$ with bounded derivatives, or $C_b^{2\beta}$), there exist constants $C_f > 0$ and $h_0 > 0$ such that

$$\big|\mathbb{E}[f(X_T)] - \mathbb{E}[f(\hat{X}_N)]\big| \;\le\; C_f\,h^{\beta} \qquad \text{for all } h \in (0, h_0]. \tag{7}$$

The weak error compares expectations of test functions, which are averages over all paths. No coupling between the discretised and the exact processes is required by the definition: $\mathbb{E}[f(\hat{X}_N)]$ depends only on the distribution of $\hat{X}_N$, regardless of how $\hat{X}_N$ is constructed.

The weak error (7) is the relevant quantity whenever the Monte Carlo estimator is used to compute an expectation: option pricing under risk-neutral measure, expected utility, the partition function in physics, and so on. In each of these settings, what is estimated by the simulation is an integral against the law of X_T, and only the marginal distribution matters.

3.4 Why the two are different

The two notions are formally different because they probe different features of the joint law of $(X_T, \hat{X}_N)$. Strong convergence sees the dependence structure between X and $\hat{X}$; weak convergence sees only their marginal distributions. The following elementary example illustrates how dramatic the difference can be.

Example 3.3 (Independent Gaussians). Let $X \sim \mathcal{N}(0,1)$ and $\hat{X} \sim \mathcal{N}(0,1)$ be independent. Then X and $\hat{X}$ have *identical* marginal distributions, so $\mathbb{E}[f(X)] = \mathbb{E}[f(\hat{X})]$ for every f and the "weak error" is zero. But the pathwise distance is large: $\mathbb{E}[|X - \hat{X}|] = \mathbb{E}[|X|] + \mathbb{E}[|\hat{X}|]$ by independence of X and $\hat{X}$ ... no, more carefully, $\mathbb{E}[|X - \hat{X}|] = \sqrt{2/\pi} \cdot \sqrt{2} = 2/\sqrt{\pi} \approx 1.13$, which is firmly bounded away from zero.

The example is a degenerate case (no time step, no SDE), but it captures the principle: the strong and weak notions can disagree arbitrarily on objects with identical marginals. For SDE schemes the difference is more subtle but operates in the same direction: weak convergence exploits cancellation across paths that strong convergence cannot see.

A useful one-way relation between the two does hold, however. Suppose X_T and $\hat{X}_N$ are coupled (run on the same Brownian path), and f is Lipschitz with constant L_f. Then by Jensen's inequality,

$$\big|\mathbb{E}[f(X_T)] - \mathbb{E}[f(\hat{X}_N)]\big| \;=\; \big|\mathbb{E}[f(X_T) - f(\hat{X}_N)]\big| \;\le\; \mathbb{E}[|f(X_T) - f(\hat{X}_N)|] \;\le\; L_f\, \mathbb{E}[|X_T - \hat{X}_N|]. \quad (8)$$

So strong order γ implies weak order $\ge \gamma$ for Lipschitz test functions. The converse fails: weak order can strictly exceed strong order, and we will see this happen for Euler–Maruyama (strong $\frac{1}{2}$, weak 1).

3.5 Which one matters in this project

For European option pricing under a risk-neutral measure, the quantity of interest is $\mathbb{E}[e^{-rT}\Phi(X_T)]$ for a payoff Φ. This is weak: marginal distributions only. *Strong convergence is irrelevant for the European pricing of Block 1.2.1 and 1.2.2.* What we require, and what we will measure empirically, is weak order.

For path-dependent pricing in Block 5, the relevant question becomes the weak error on functionals of the entire path $\mathbb{E}[\Phi(X_{[0,T]})]$, not just on $\mathbb{E}[\Phi(X_T)]$. The marginals at the grid points are not enough; in particular, for barrier options the discretised supremum has its own bias of order $\sqrt{h}$ that adds to any scheme's marginal error, and a Brownian-bridge correction will be deployed. Strong convergence remains irrelevant to barrier pricing *per se*.

For pathwise sensitivity analysis in Block 4, by contrast, strong convergence becomes essential: the pathwise estimator differentiates the simulated path with respect to a parameter, and that differentiation requires the path to be a faithful pathwise approximation of the true process.

We therefore care about both notions across the project, but in different blocks. For Block 1.2 specifically the operational conclusion is that we will measure both orders empirically (the strong order is interesting because it is the surprising one: $\frac{1}{2}$ for Euler), but only the weak order is the metric that justifies using a given scheme for European pricing.

4 The Itô-Taylor expansion

This section develops the device by which discretisation schemes for SDEs are derived systematically. The mechanics generalise the deterministic Taylor expansion, with the role of the chain rule played by Itô's formula. The treatment is detailed because the expansion is the principal source of conceptual novelty in this block.

4.1 Reminder: Taylor expansion for ODEs

Consider the deterministic ODE $\dot{x} = a(x)$. The Taylor expansion of $x(t+h)$ around $x(t)$ to second order reads

$$x(t+h) \;=\; x(t) + h\,\dot{x}(t) + \tfrac{1}{2}h^2\ddot{x}(t) + O(h^3). \tag{9}$$

Using the ODE itself, $\dot{x} = a(x)$ and $\ddot{x} = a'(x)\,\dot{x} = a'(x)\,a(x)$. Substituting,

$$x(t+h) \;=\; x(t) + h\,a(x(t)) + \tfrac{1}{2}h^2 a'(x(t))\,a(x(t)) + O(h^3). \tag{10}$$

The explicit Euler scheme for ODEs is precisely the truncation of (10) after the first-order term: $x_{n+1} = x_n + h\,a(x_n)$. Higher-order schemes (improved Euler, Runge–Kutta) keep more terms.

This logic extends to SDEs but with two complications: (i) the solution X_t of (2) is not differentiable in t, so we cannot literally Taylor-expand in time; (ii) the chain rule for $f(X_t)$ is Itô's formula, which carries a quadratic-variation term absent in the ODE case. The first difficulty is overcome by working with the integral form of the SDE, the second by accepting Itô's formula as the SDE chain rule.

4.2 Itô's formula and the generator

Recall Itô's formula for a C^2 function f:

$$\mathrm{d}f(X_t) \;=\; \mathcal{L}f(X_t)\,\mathrm{d}t \;+\; f'(X_t)\,b(X_t)\,\mathrm{d}W_t, \tag{11}$$

where the *Itô generator* $\mathcal{L}$ is the operator

$$\mathcal{L}f(x) \;\stackrel{\text{def}}{=}\; a(x)f'(x) + \tfrac{1}{2}b^2(x)f''(x). \tag{12}$$

Integrating (11) from t to $t+h$,

$$f(X_{t+h}) \;=\; f(X_t) + \int_t^{t+h} \mathcal{L}f(X_s)\,\mathrm{d}s \;+\; \int_t^{t+h} f'(X_s)\,b(X_s)\,\mathrm{d}W_s. \tag{13}$$

Two interpretations of $\mathcal{L}$ are useful. As a differential operator, $\mathcal{L}$ is the second-order operator that gives the drift of $f(X)$ under (2); the second-derivative term comes from the quadratic variation of W. As an analogue of $a \cdot f'$ in the deterministic case, $\mathcal{L}$ plays the role of the "time derivative" operator for SDEs: it is the Markov generator of the diffusion X, and $\mathcal{L}f(x) = \lim_{h\to 0} h^{-1}\left(\mathbb{E}[f(X_{t+h})|X_t = x] - f(x)\right)$.

The integral identity (13) is the SDE analogue of the fundamental theorem of calculus. The Itô-Taylor expansion is what results from *iterating* this identity.

4.3 First iteration: the SDE in integral form

Apply (13) with $f(x) = x$. Since $f' = 1$ and $f'' = 0$, the generator gives $\mathcal{L}f = a$, and $f' \cdot b = b$. The result is

$$X_{t+h} \;=\; X_t + \int_t^{t+h} a(X_s)\,\mathrm{d}s \;+\; \int_t^{t+h} b(X_s)\,\mathrm{d}W_s. \tag{14}$$

This is just the SDE (2) in integral form, no surprise. What it gives us, however, is a starting point: X_{t+h} is expressed in terms of X_t plus two integrals over $[t, t+h]$ involving $a(X_s)$ and $b(X_s)$. If we could replace $a(X_s)$ and $b(X_s)$ by $a(X_t)$ and $b(X_t)$ at the cost of a small error, we would have

$$X_{t+h} \;\approx\; X_t + a(X_t)h + b(X_t)\Delta W, \tag{15}$$

which is the Euler–Maruyama scheme. The question is what "small error" means and how to quantify it. We answer this by applying (13) again, this time to a and to b.

4.4 Second iteration: expanding the integrands

Apply (13) with $f = a$:

$$a(X_s) \;=\; a(X_t) + \int_t^s \mathcal{L}a(X_u)\,\mathrm{d}u \;+\; \int_t^s a'(X_u)\,b(X_u)\,\mathrm{d}W_u \qquad \text{for } s \in [t, t+h], \tag{16}$$

and similarly with $f = b$:

$$b(X_s) \;=\; b(X_t) + \int_t^s \mathcal{L}b(X_u)\,\mathrm{d}u \;+\; \int_t^s b'(X_u)\,b(X_u)\,\mathrm{d}W_u. \tag{17}$$

Substitute (16) and (17) into (14). The first integral becomes

$$\int_t^{t+h} a(X_s)\,\mathrm{d}s \;=\; a(X_t)\int_t^{t+h} \mathrm{d}s \;+\; \int_t^{t+h}\!\!\int_t^s \mathcal{L}a(X_u)\,\mathrm{d}u\,\mathrm{d}s \;+\; \int_t^{t+h}\!\!\int_t^s a'(X_u)b(X_u)\,\mathrm{d}W_u\,\mathrm{d}s. \tag{18}$$

The second integral becomes

$$\int_t^{t+h} b(X_s)\,\mathrm{d}W_s \;=\; b(X_t)\int_t^{t+h} \mathrm{d}W_s + \int_t^{t+h}\!\!\int_t^s \mathcal{L}b(X_u)\,\mathrm{d}u\,\mathrm{d}W_s + \int_t^{t+h}\!\!\int_t^s b'(X_u)b(X_u)\,\mathrm{d}W_u\,\mathrm{d}W_s. \tag{19}$$

Combining,

$$X_{t+h} \;=\; X_t \;+\; a(X_t)h \;+\; b(X_t)\Delta W \;+\; b(X_t)b'(X_t)\,I_{(1,1)} \;+\; R, \tag{20}$$

where:

- $\Delta W = W_{t+h} - W_t = \int_t^{t+h} \mathrm{d}W_s$,
- $I_{(1,1)} \stackrel{\text{def}}{=} \int_t^{t+h}\!\int_t^s \mathrm{d}W_u\,\mathrm{d}W_s$ is the iterated Itô integral over two Brownian increments,
- R collects all remaining terms: the drift–drift double integral $\int\!\int \mathcal{L}a$, the drift–diffusion mixed $\int\!\int a'b$, and so on, plus the leftover integrand factors that we replaced by their values at t.

We have deliberately written (20) so that the $b\,b' \cdot I_{(1,1)}$ term is isolated, because it is the term that distinguishes Milstein from Euler.

4.5 Closed form for the iterated integral

The iterated Itô integral $I_{(1,1)}$ admits a closed form. Apply Itô's formula to $f(W_t) = \frac{1}{2}W_t^2$:

$$\mathrm{d}\big(\tfrac{1}{2}W_t^2\big) \;=\; W_t\,\mathrm{d}W_t + \tfrac{1}{2}\,\mathrm{d}t, \tag{21}$$

which integrated from t to $t+h$ gives

$$\tfrac{1}{2}(W_{t+h}^2 - W_t^2) \;=\; \int_t^{t+h} W_s\,\mathrm{d}W_s \;+\; \tfrac{1}{2}h. \tag{22}$$

Subtract $W_t \int_t^{t+h} \mathrm{d}W_s = W_t\Delta W$ from both sides and use $W_s - W_t = \int_t^s \mathrm{d}W_u$ for $s \in [t, t+h]$:

$$\int_t^{t+h} (W_s - W_t)\,\mathrm{d}W_s \;=\; \tfrac{1}{2}((W_{t+h} - W_t)^2 - h) \;=\; \tfrac{1}{2}\big((\Delta W)^2 - h\big). \tag{23}$$

The left-hand side is exactly $I_{(1,1)}$ (using $W_s - W_t = \int_t^s \mathrm{d}W_u$ and Fubini for stochastic integrals, which holds in this Itô setting). Therefore

$$I_{(1,1)} \;=\; \tfrac{1}{2}\big((\Delta W)^2 - h\big). \tag{24}$$

This identity is the key technical ingredient of the Milstein scheme: it expresses the iterated integral, which is in principle a σ-algebra-rich object built from the Brownian increment plus its variance structure, in terms of the Brownian increment alone. We do not need to simulate $I_{(1,1)}$ separately; given $\Delta W \sim \mathcal{N}(0,h)$, we already know $I_{(1,1)}$ exactly.

Remark 4.1 (Multidimensional case). For SDEs driven by several independent Brownian motions $W^{(1)}, \dots, W^{(d)}$, the analogue of (24) for cross-iterated integrals such as $\int\int \mathrm{d}W^{(i)}\,\mathrm{d}W^{(j)}$ with $i \neq j$ *does not* have a closed form in terms of the increments. Simulating these cross-iterated integrals exactly requires extra work (or a Brownian bridge construction), and this is the reason the Milstein scheme in dimension $d \geq 2$ is significantly more demanding than in dimension 1. Phase 2 stays in the scalar setting, so (24) is sufficient.

4.6 The expansion to second order

Substituting (24) into (20):

$$X_{t+h} \;=\; X_t + a(X_t)h + b(X_t)\Delta W \;+\; \tfrac{1}{2}b(X_t)b'(X_t)\left[(\Delta W)^2 - h\right] \;+\; R. \tag{25}$$

The remainder R contains all the terms that have not been written explicitly: the double drift integral, the mixed integrals, and the contributions from the integrands $\mathcal{L}a, a'b, \mathcal{L}b, b'b$ evaluated at points X_u rather than X_t. Each of these terms is, on closer inspection, of higher order in h than the explicit terms: the double drift integral is $O(h^2)$, the mixed drift–diffusion integrals are $O(h^{3/2})$ in L^1, and so on. The next subsection makes this precise informally.

4.7 Sizes of stochastic increments

A useful heuristic for predicting which terms matter at which order is the rule of thumb

$$\Delta W \;\sim\; \sqrt{h}, \qquad h \;\sim\; h. \tag{26}$$

That is, the random increment ΔW has standard deviation $\sqrt{h}$, so its "typical size" in any norm sensitive to it is $\sqrt{h}$, while a deterministic increment of size h stays h. Combinations:

- $(\Delta W)^2 \sim h$ (since $\mathrm{Var}(\Delta W) = h$),
- $\Delta W \cdot h \sim h^{3/2}$,
- $(\Delta W)^3 \sim h^{3/2}$ (third moment of $\mathcal{N}(0,h)$ is 0, but typical magnitudes are $h^{3/2}$),
- double iterated integrals like $\int\int \mathrm{d}W\,\mathrm{d}W \sim h$ (this is precisely $I_{(1,1)}$ from (24)),
- double iterated integrals like $\int\int \mathrm{d}s\,\mathrm{d}s \sim h^2$.

This calculus is informal but accurate enough to predict the orders of all schemes we will encounter. With it, the structure of (25) becomes:

- $a(X_t)\,h$: order h.
- $b(X_t)\,\Delta W$: order $h^{1/2}$.
- $\frac{1}{2}b\,b'\,[(\Delta W)^2 - h]$: order h.
- R (terms not written): order $h^{3/2}$ and higher.

5 The Euler–Maruyama scheme

5.1 Derivation as first-order truncation

The Euler–Maruyama scheme is obtained by truncating (25) after the ΔW term: keep h and $h^{1/2}$, drop h from $I_{(1,1)}$ and everything beyond. The recursion is

$$\hat{X}_{n+1} \;=\; \hat{X}_n \;+\; a(\hat{X}_n)\,h \;+\; b(\hat{X}_n)\,\Delta W_n, \qquad \hat{X}_0 = x_0. \tag{27}$$

This is the natural extension to SDEs of the explicit Euler scheme for ODEs: replace a by $b\,dW$ where appropriate and add the stochastic increment.

The truncation drops, at each step, a quantity that (25) reveals to be of size h (the $I_{(1,1)}$-related term) plus higher-order remainders.

5.2 Convergence orders

Theorem 5.1 (Convergence of Euler–Maruyama)**.** *Suppose $a, b \in C^2(\mathbb{R})$ with a, b, a', b' Lipschitz and bounded. The Euler–Maruyama scheme* (27) *converges to the solution of* (2) *with strong order $\gamma = \frac{1}{2}$ and weak order $\beta = 1$.*

The proof is technical and is given in Kloeden–Platen, § 10.2 (strong) and § 14.5 (weak). We do not reproduce it. What we do is explain, at the level of intuition, why the orders are $\frac{1}{2}$ and 1 respectively.

Why strong order $\frac{1}{2}$. The local truncation error of one Euler step is the dropped term $\frac{1}{2}bb'[(\Delta W)^2 - h] + O(h^{3/2})$. In L^1, $|(\Delta W)^2 - h|$ has expectation $\mathbb{E}[|(\Delta W)^2 - h|] \sim h$ (this is a moment of a centred chi-square), so the local strong error is $O(h)$. There are $N = T/h$ steps and the errors combine in L^1 via a Gronwall-type argument that is essentially a square-root law (the variance accumulates linearly, not the standard deviation). The global strong error is thus $\sqrt{N \cdot h^2} = \sqrt{Th}$, that is, $O(h^{1/2})$.

Why weak order 1. The local truncation error has expectation $\frac{1}{2}\mathbb{E}[bb' \cdot ((\Delta W)^2 - h)]$, which *vanishes* because $\mathbb{E}[(\Delta W)^2 - h] = 0$. The leading contribution to the weak error therefore comes from the next term in the expansion, which is $O(h^{3/2})$ but again has zero mean to leading order; what survives are second-moment terms of size $O(h^2)$ locally, accumulating to $O(h)$ globally. Hence weak order 1.

The asymmetry $\gamma = \frac{1}{2}, \beta = 1$ is therefore not an accident: it reflects the fact that the dominant "missing" term $\frac{1}{2}bb'[(\Delta W)^2 - h]$ is path-dependent (so seen by L^1) but mean-zero (so invisible to $\mathbb{E}[\cdot]$).

5.3 What the order means in practice

Suppose we want to halve the strong error: we must *quarter* the step size h (since strong error $\sim h^{1/2}$). The number of steps $N = T/h$ quadruples and so does the simulation cost (per path). To halve the weak error: we must halve the step size; cost only doubles.

For European pricing, where weak order is what matters, Euler is respectable. For pathwise applications, the strong order $\frac{1}{2}$ becomes a real bottleneck and is the motivation for moving to Milstein.

5.4 Application to GBM

For geometric Brownian motion, $a(x) = rx$ and $b(x) = \sigma x$, so (27) becomes

$$\hat{S}_{n+1} \;=\; \hat{S}_n + r\,\hat{S}_n\,h \;+\; \sigma\,\hat{S}_n\,\Delta W_n \;=\; \hat{S}_n\big(1 + rh + \sigma\Delta W_n\big). \tag{28}$$

The recursion is multiplicative, which is convenient. A nuisance, specific to GBM under Euler: there is no a priori guarantee that $\hat{S}_n$ remains positive: if a ΔW_n is negative enough that $1 + rh + \sigma\Delta W_n < 0$, the simulated price becomes negative or zero, which is unphysical. The probability of this for realistic parameters and reasonable h is small but nonzero, and the log-Euler scheme (apply Euler in $\log S$, then exponentiate) sidesteps the issue. We will mention this in the implementation but not adopt it: the standard Euler is the canonical scheme and exhibits the canonical convergence behaviour we want to validate.

6 The Milstein scheme

6.1 Derivation as second-order truncation

The Milstein scheme is obtained by keeping the $I_{(1,1)}$ term in (25), that is, truncating after order h:

$$\hat{X}_{n+1} \;=\; \hat{X}_n \;+\; a(\hat{X}_n)\,h \;+\; b(\hat{X}_n)\,\Delta W_n \;+\; \tfrac{1}{2}b(\hat{X}_n)\,b'(\hat{X}_n)\,\big[(\Delta W_n)^2 - h\big], \qquad \hat{X}_0 = x_0. \quad (29)$$

The new term, called the Milstein correction, is the iterated Itô integral $I_{(1,1)}$ multiplied by $b\,b'$ evaluated at the current state. By (24) this term is computable directly from ΔW_n and h, with no extra random variables. The cost of Milstein over Euler is the evaluation of $b'(\hat{X}_n)$ and a few extra arithmetic operations.

6.2 Convergence orders

Theorem 6.1 (Convergence of Milstein)**.** *Suppose $a, b \in C^2(\mathbb{R})$ with a, b, a', b', a'', b'' Lipschitz and bounded. The Milstein scheme* (29) *converges to the solution of* (2) *with strong order $\gamma = 1$ and weak order $\beta = 1$.*

Why strong order 1**.** By including the $I_{(1,1)}$ term, Milstein eliminates the $O(h)$ local strong error of Euler. The next remainder is $O(h^{3/2})$, which accumulates over $N = T/h$ steps to $O(h)$ globally. Hence strong order 1.

Why weak order remains 1**.** Adding a mean-zero correction to Euler does not improve the weak error: the leading non-zero expectation term is unchanged. To get weak order > 1, one needs to correct further moments of the increment; this requires schemes like the order-2 weak scheme of Platen, which is genuinely more involved and beyond the scope of this block.

6.3 Application to GBM

For GBM, $b(x) = \sigma x$, so $b'(x) = \sigma$ and $b(x)b'(x) = \sigma^2 x$. The recursion becomes

$$\hat{S}_{n+1} \;=\; \hat{S}_n + r\hat{S}_n h + \sigma\hat{S}_n\Delta W_n + \tfrac{1}{2}\sigma^2\hat{S}_n\big[(\Delta W_n)^2 - h\big]. \quad (30)$$

Again multiplicative, again with a positivity issue (although the correction term raises the lower bound: the squared term $(\Delta W_n)^2$ is always nonnegative, so the correction is bounded below by $-\frac{1}{2}\sigma^2\hat{S}_n h$, and the worst-case loss per step is mitigated). We will report empirically what fraction of paths, if any, develop nonpositive prices in our test parameters.

7 Practical considerations and which scheme to use

The textbook recommendation can be misleading and deserves an explicit unpacking. Phrases like "Milstein is more accurate than Euler" are true in the strong sense and *false* in the weak sense, where the two schemes are tied at order 1.

7.1 For European option pricing

We compute $\mathbb{E}[e^{-rT}\Phi(S_T)]$. This is a marginal expectation, the weak setting. Both Euler and Milstein have weak order 1. Milstein costs more per step (one extra multiplication and one derivative evaluation; the latter can be analytically symbolic and hence "free" if the model coefficients are simple). The only benefit of Milstein over Euler in this setting is a smaller constant in front of the h asymptote, not an improved order.

In practice for pricing under GBM (and for many local volatility models), the constant improvement is too small to justify the extra machinery: Euler with a slightly smaller h achieves the same weak error at lower or comparable cost. Block 1.2.1 will demonstrate empirically that for European pricing under GBM, Euler at $h = T/200$ and Milstein at $h = T/100$ (matched cost) deliver comparable accuracy.

7.2 For pathwise sensitivity (Block 4)

The pathwise method estimates greeks by differentiating the simulated path with respect to a parameter, then averaging. For the differentiation to be meaningful, the simulated path must be a faithful pathwise approximation, which is the strong setting. Milstein's strong order 1 versus Euler's $\frac{1}{2}$ becomes a genuine advantage: Milstein needs $\sqrt{h}$ as many steps as Euler to attain the same strong error, an asymptotic factor that grows without bound as $h \to 0$.

7.3 For path-dependent payoffs (Block 5)

The expectation is now over the path, $\mathbb{E}[\Phi(S_{[0,T]})]$. Two sources of error: the marginal scheme error (weak, both schemes $O(h)$) and the discretisation of the payoff functional itself (barriers, suprema). The latter dominates for barrier options at $O(\sqrt{h})$, regardless of scheme. So Euler is sufficient here, and the relevant improvement is in the payoff treatment (Brownian bridge correction), not in the scheme.

7.4 Summary

The verdict is unambiguous: *for the European pricing of Block 1.2, Euler–Maruyama is the right choice and Milstein is implemented for completeness and as a comparison baseline rather than because it offers any operational advantage.* Block 1.2.1 will implement Euler as the working pricer; Block 1.2.2 will implement Milstein as a side-by-side benchmark. The empirical convergence study at the heart of validation will measure both strong and weak orders for both schemes, confirming the predictions of Theorems 5.1 and 6.1.

8 Roadmap for Blocks 1.2.1 and 1.2.2

Block 1.2.1 will implement Euler–Maruyama for GBM and validate it empirically. The validation has three components:

1. **Strong-order verification**: simulate $\hat{S}_T$ on the *same* Brownian path as the exact S_T (the latter computed by the closed-form formula evaluated at $W_T = \sum_n \Delta W_n$), measure $\mathbb{E}[|S_T - \hat{S}_T|]$ across many paths, plot in log-log against h, fit a line, expect slope $\frac{1}{2}$.

2. **Weak-order verification**: with $\Phi(s) = (s-K)^+ e^{-rT}$, measure $|\mathbb{E}[\Phi(S_T)] - \mathbb{E}[\Phi(\hat{S}_T)]|$ across many paths and many h, plot in log-log, expect slope 1.

3. **Sanity checks**: containment of the BS price by the MC CI for moderate h, monotonicities, reduction to Block 1.1's exact sampler in the limit $h \to 0$ at fixed n.

The first two are direct empirical confirmations of the convergence theorems; the third anchors the implementation in the existing infrastructure.

Block 1.2.2 will implement Milstein under the same protocol. The strong order should change from $\frac{1}{2}$ to 1; the weak order should remain 1, matching Euler. This is the canonical demonstration of the asymmetry between the two notions of convergence.

Architecturally, both schemes will be added to the existing `quantlib.monte_carlo` (Python) and `quant/monte_carlo` (C++) modules, alongside the exact GBM sampler from Block 1.1. The `mc_estimator` reducer is unchanged. The pricer wrappers (`mc_european_call_euler`, `mc_european_call_milstein`) will be thin orchestrators on top of new path samplers.

9 Bibliographic notes

The canonical reference for the numerical analysis of SDEs is Kloeden & Platen [3], especially Chapters 5 (stochastic Taylor expansions), 9–10 (Euler–Maruyama and strong convergence), and 14–15 (weak convergence and Milstein-type schemes). The treatment is comprehensive and self-contained but demanding; the relevant theorems are stated with multiple regularity hypotheses that we have stylised in Theorems 5.1 and 6.1. For a reader-friendlier introduction, Higham [2] is a classic survey article that develops exactly the empirical convergence demonstration we will reproduce in the implementation. Glasserman [1, Chapter 6] treats the topic from the financial Monte Carlo angle and includes discussion of the practical relevance of weak versus strong convergence in the contexts we will encounter in subsequent blocks.

References

[1] P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2004.

[2] D. J. Higham, *An algorithmic introduction to numerical simulation of stochastic differential equations*, SIAM Review, 43(3):525–546, 2001.

[3] P. E. Kloeden and E. Platen, *Numerical Solution of Stochastic Differential Equations*, Springer, 1992.

[4] A. Etheridge, *A Course in Financial Calculus*, Cambridge University Press, 2002.